GENERAL STAFF OF THE ARMED FORCES OF THE RUSSIAN FEDERATION

Directory
FPV Drones Used by the Russian Armed Forces in the SMO



Vladikavkaz 2024

This reference book presents FPV drones and countermeasures used in various conditions by the Armed Forces of the Russian Federation in special military operations.

An analysis of the use of UAVs in special military operations showed that the total share of FPV drones is 90% of all types of UAVs. This is due to the high speed and maneuverability, continuous improvement of manufacturing technologies, and low cost of FPV drones compared to other UAVs.

This reference book was prepared at for the purpose of disseminating information about FPV drones used in the Armed Forces, and is intended for officials of military units, central military command bodies, military educational institutions, and research organizations of the Ministry of Defense of the Russian Federation.

The reference book was prepared by a team of authors from the department (generalization of combat experience) of the OGV(s) by Colonels Rumyantsev V.V., Colonel S.A. Torbin, Lieutenant Colonels K.A. Osokin and V.V. Shepranov, Majors D.V. Emelyanov, A.A. Mosolov, A.D. Pobegailo, and I.V. Shatalov.

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Table of contents

Section 1

FPV DRONES
COPTER TYPE



PAGE 4

Section 2

FPV DRONES
AIRPLANE
AND HELICOPTER TYPE



PAGE 31

Section 3

COPTER-TYPE UAV WITH
FPV FUNCTION



PAGE 35

Section 4

IMPACT LOAD OF FPV
DRONES



PAGE 45

Section 5

MEANS OF SUPPRESSING
FPV DRONES



PAGE 51

Purpose, main tasks, design features and methods of combating FPV drones

FPV drones are designed for hitting objects, living forces, armored, light armored and automotive equipment of the enemy, ensuring situational awareness during combat operations.

The main tasks are:

- reconnaissance of terrain, enemy targets, and routes of movement of personnel and armored vehicles;
- fire damage to personnel, armored, lightly armored, and motorized vehicles;
- fire support for attacking and defending troops;
- interception of enemy drones (airborne destruction);
- remote mining.

The design features of FPV drones, **as opposed** to quadcopter-type UAVs, are:

- the presence of video goggles and a control panel in "first-person view" mode;
- no satellite navigation receiver or autopilot;
- the ability to use transceiver control modules operating at different frequencies;
- the use of specialized, adapted ammunition and homemade explosive devices.

Specialized ammunition is factory-made ammunition designed for use with this type of UAV.

Adapted – ammunition and warheads for grenade launcher systems, engineering weapons, and ammunition for deployment from UAVs, modified in technical units for use with FPV drones.

Homemade – explosive devices manufactured in technical units. Various types of explosives, chemical mixtures, fuses, and striking elements may be used in their manufacture.

use various types of explosives, chemical mixtures, detonators, and striking elements.

Combating enemy FPV drones is divided into two main components: detection and counteraction. Methods of combating enemy FPV drones are divided into active and passive.

Active methods include:

- destruction of FPV drones with small arms;
- radio-electronic suppression of FPV drone control signals;
- Fire damage to enemy FPV control points.

Passive methods of combat include:

- building fortifications;
- creating false positions;
- camouflaging one's positions;
- use of protective nets and metal structures

Section 1
FPV COPTER-TYPE DRONES

- FPV drone "RTD-K"
- FPV drone "Kort"
- 5-inch FPV drone
- 7-inch FPV drone
- FPV drone "Darwin X9"
- FPV drone "Boomerang"
- FPV drone "Brus-1"
- FPV drone "Boomerang-ERA"
- FPV drone "Hortensia-7"
- FPV drone "Hortensia-10"
- FPV drone "XL-10"
- FPV drone "Ploschad-K"
- FPV drone "PVC-1"
- FPV drone "Chimera"
- FPV drone "Ovod"
- FPV drone "ASK X-2"
- FPV drone "Shmel"
- FPV drone "Veles"
- FPV drone "D-11-1"
- FPV drone "Kurs-800TB"
- FPV drone "Zhmyil-17"
- FPV drone "Vespa"
- FPV drone "Skorets-7"
- FPV drone "Skvorets-10"
- FPV drone "Skorets-Z"
- FPV drone "Skorets PRO"
- FPV drone "Predator-2PG"
- FPV drone "Katyusha"
- FPV drone "Product 1-390"
- FPV drone "Slepen"
- FPV drone "Ryabchik"
- FPV drone "Courier"
- FPV drone A8 3300 "Alyosha"
- FPV drone "Boomerang-ERA"
- FPV drone "Granat-3"
- FPV drone "Monolith-FPV-7"
- FPV drone "Osa"
- FPV drone "Loon-7 (10)"
- FPV drone "Veter"
- Antonov FPV drone
- 9-inch FPV drone
- 10-inch FPV drone
- FPV drone "Komar"
- FPV drone "Ultimatum"
- FPV drone "Piranha-7"
- FPV drone "Piranha-10"
- FPV drone "Lutik-Kamikaze"
- FPV drone "Boomerang-10"
- FPV drone "Type-450"
- FPV drone "Microbe"
- FPV drone "Dozor"
- FPV drone "Upyr"

FPV drone "RTD-K"

The RTD-K FPV drone is designed for high-precision strikes against enemy air defense systems and manpower, with manual targeting via video signal.



Parameter name	Value
Maximum takeoff weight, kg	6
Payload, kg	up to 2.5
Range, km	up to 6
Flight time, min	8 / 25
Speed, km/h	170
Service ceiling, m	2000
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	930 / 2400; 5800
Manufacturer	NPO Rustechnodron

FPV drone "Court"

FPV drone "Kort" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum take-off weight, kg	6
Payload weight, kg	up to 2.5
Range, km	up to 6
Flight time, min	6 / 25
Speed, km/h	150
Service ceiling, m	3000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4000
Operating frequency range, MHz	433–915/1200
Manufacturer	TSUGAM LLC

5-inch FPV drone

FPV drone 5 inches is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	3.4
Payload, kg	up to 1.4
Range, km	up to 5
Flight time, min	up to 10
Speed, km/h	up to 110
Practical ceiling, m	100
Operating temperature range, °C	from -10 to +30
Start/landing method	vertical
Battery capacity, mAh	3600
Operating frequency range, MHz	1080–1320
Manufacturer	Russia

7-inch FPV drone

FPV drone 7 inches designed for strike and reconnaissance operations, support actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	4.8
Payload, kg	up to 3
Range, km	up to 18
Flight time, min	up to 15
Speed, km/h	up to 100
Practical ceiling, m	140
Operating temperature range, °C	from -20 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	5800
Operating frequency range, MHz	1080–1320
Manufacturer	Russia

Darwin X9 FPV drone

FPV drone Darwin X9 is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	5
Payload, kg	up to 3.7
Range, km	up to 3
Flight time, min	up to 10
Speed, km/h	up to 170
Service ceiling, m	6000
Operating temperature range, °C	from -10 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	5800 / 915; 2400
Manufacturer	Russia

FPV drone "Boomerang"

FRV drone "Boomerang" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	5
Payload weight, kg	up to 3.5
Range, km	up to 4.5
Flight time, min	up to
Speed, km/h	up to 180
Practical ceiling, m	1000
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	433; 868; 915 / 5800
Manufacturer	Russia

FPV drone "Brus-1"

FPV drone "Brus-1" is designed for strike and reconnaissance operations, escort assault groups



Parameter name	Value
Maximum takeoff weight, kg	3.5
Payload, kg	up to 1.5
Range, km	up to 4
Flight time, min	up to 25
Speed, km/h	up to 150
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4000
Operating frequency range, MHz	868/1200; 5800
Manufacturer	TSUGAM LLC

FPV drone "Boomerang-E"

The FRV drone "Boomerang-ERA" is designed to destroy enemy manpower and military equipment



Parameter name	Value
Maximum takeoff weight, kg	5.8
Payload weight, kg	up to 4
Range, km	up to 10
Flight time, min	up to 10
Speed, km/h	up to 165
Service ceiling, m	3000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868–950 / 5333–5945
Manufacturer	LLC "NPO Rustekhdrone

FPV drone "Hortensia-7"

FRV drone "Hortensia-7" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	3.5
Payload, kg	up to 2
Range, km	up to 6
Flight time, min	up to 8
Speed, km/h	up to 130
Practical ceiling, m	1600
Operating temperature range, °C	from -10 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	1080–1320
Manufacturer	KB LiS LLC Vsevolozhsk

FPV drone "Hortensia-10"

FRV drone "Hortensia-10" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	5
Payload, kg	up to 3
Range, km	up to 8
Flight time, min	up to 8
Speed, km/h	up to 150
Service ceiling, m	1600
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	1080–1320
Manufacturer	KB LiS LLC Vsevolozhsk

FPV drone XL-10

FPV drone "XL-10" is designed for strike and reconnaissance operations, escort actions assault teams



Parameter name	Value
Maximum takeoff weight, kg	4.7
Payload, kg	up to 3.5
Range, km	up to 10
Flight time, min	up to 12
Speed, km/h	up to 140
Practical ceiling, m	1000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868; 915 / 2400; 5800
Manufacturer	Russia

FPV drone "Ploschad-K"

FPV drone "Ploshchad-K" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	2
Payload, kg	up to 1
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 125
Practical ceiling, m	1000
Operating temperature range, °C	-10 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868; 915 / 5850
Manufacturer	Ploshchad JSC Moscow

FPV drone "PVC-1"

FPV drone "PVC-1" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	4.2
Payload, kg	up to 3
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 180
Service ceiling, m	800
Operating temperature range, °C	-10 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	868; 915 / 5800
Manufacturer	Rustakt LLC, Rostov-on-Don

FPV drone "Chimera"

FPV drone "Chimera" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	4.2
Payload, kg	up to 3
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 180
Service ceiling, m	800
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	433; 868; 915 / 5800
Manufacturer	Russia

FPV drone "Ovod"

FPV drone "Ovod" is designed for strike and reconnaissance operations, escort operations assault teams



Parameter name	Value
Maximum takeoff weight, kg	5.4
Payload, kg	up to 4.5
Range, km	up to 5
Flight time, min	up to 10
Speed, km/h	up to 180
Practical ceiling, m	5000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	385 / 4990 – 5200
Manufacturer	RLVT LLC Tula

FPV drone "ASK X-2"

FRV drone ASK X-2 is designed for strike and reconnaissance operations, escort actions assault teams



Parameter name	Value
Maximum take-off weight, kg	4.8
Payload, kg	up to 2.8
Range, km	up to 15
Flight time, min	up to 30
Speed, km/h	up to 70
Practical ceiling, m	3000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	900 / 2400; 5800
Manufacturer	ASK LLC Moscow

FPV drone "Shmel"

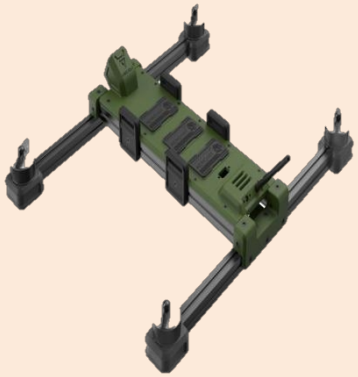
FPV drone "Bumblebee" is designed for strike and reconnaissance operations, escort actions assault teams



Parameter name	Value
Maximum takeoff weight, kg	2
Payload, kg	up to 1.5
Range, km	up to 5000 / up to 13000 (when using external antenna)
Flight time, min	up to 15
Speed, km/h	up to 150
Practical ceiling, m	2000
Operating temperature range, °C	from -30 to +50
Takeoff/landing method	Vertical
Battery capacity, mAh	4200
Operating frequency range, MHz	1080–1320
Manufacturer	IP Zhuchenko S.V. Rostov-on-Don

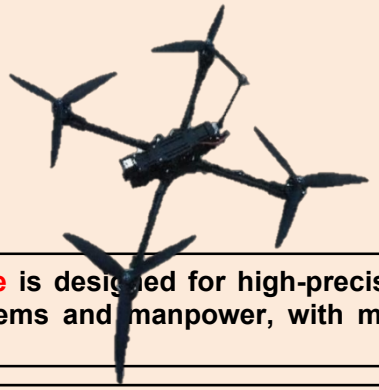
FPV drone "Veles"

FRV drone "Veles" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	4.3
Payload, kg	up to 3
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 100
Practical ceiling, m	1000
Operating temperature range, °C	From -10 to +50
Start/landing method	Vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	1200; 2400; 5800
Manufacturer	Russia

FPV drone "D-11-1"



The D-11-1 FPV drone is designed for high-precision strikes against enemy air defense systems and manpower, with manual targeting via video signal.

Parameter name	Value
Maximum take-off weight, kg	4.7
Payload weight, kg	up to 4
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 120
Practical ceiling, m	3000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868; 915 / 1200; 5800
Manufacturer	IE Zhuchenko S.V. Rostov-on-Don

FPV drone "Kurs-800TB"

The Kurs-800TB FPV drone is designed for reconnaissance, determining coordinates at night and during the day, as well as for striking identified targets.



Parameter name	Value
Maximum take-off weight, kg	2.5
Payload, kg	up to 0.8
Range, km	up to 6
Flight time, min	up to 50
Speed, km/h	up to 36
Practical ceiling height, m	3000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	7000
Operating frequency range, MHz	868; 915 / 5800
Manufacturer	JSC "OKB ASTRON" Moscow Region

FPV drone "Zhml-17"

FPV drone Zhmyil-17 is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	2.2
Payload, kg	1.4
Range, km	up to 8
Flight time, min	up to 15
Speed, km/h	up to 130
Practical ceiling, m	5000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4000
Operating frequency range, MHz	868–915 / 5725–5850
Manufacturer	MOLOT ARMS LLC

FPV drone "Vespa"

FPV drone "Vespa" is designed for strike and reconnaissance operations, escort actions assault teams



Parameter name	Value
Maximum takeoff weight, kg	2.5
Payload weight, kg	up to 1
Range, km	up to 15
Flight time, min	up to 15
Speed, km/h	up to 35
Practical ceiling, m	1000
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868–915 / 5800
Manufacturer	Prom Composite LLC, Moscow

FPV drone Skvorets-7

FPV drone "Skvorets-7" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	5.3
Payload, kg	up to 3.5
Range, km	up to 20
Flight time, min	up to 20
Speed, km/h	up to 150
Practical ceiling, m	800
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	300 / 4900 – 5100
Manufacturer	Technodron

FPV drone "Skvorets-10"

FPV drone "Skvorets-10" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	6.3
Payload mass, kg	up to 3.5
Range, km	up to 30
Flight time, min	up to 15
Speed, km/h	up to 150
Practical ceiling, m	1000
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	7000
Operating frequency range, MHz	300–500 / 4900–5100
Manufacturer	Technodron

FPV drone "Skvorets-Z"

FPV drone "Skorets-Z" is designed for to defeat the enemy for moving and stationary targets in automatic mode objects.



Parameter name	Value
Maximum takeoff weight, kg	4
Payload weight, kg	up to 2.5
Range, km	up to 10
Flight time, min	up to 20
Speed, km/h	up to 100
Service ceiling, m	800
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	7000
Operating frequency range, MHz	300 / 4900 – 5100
Manufacturer	Technodron

FPV drone "Skvorets PRO"

The Skvorets PRO FPV drone is designed for reconnaissance and high-precision strikes against enemy forces. It does not require advanced piloting skills, automatically maintains altitude, and autonomously flies to the target.



Parameter name	Value
Maximum takeoff weight, kg	3
Payload mass, kg	up to 1.5
Range, km	up to 10
Flight time, min	up to 20
Speed, km/h	up to 108
Service ceiling, m	800
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	300 / 4900 – 5100
Manufacturer	Technodron

FPV drone "Predator-2PG"

The FPV armored "Predator-2PG" is designed for the enemy with PG-7S hand RPG-7



Parameter name	Value
Maximum takeoff weight, kg	6.5
Payload, kg	up to 5
Range, km	up to 7
Flight time, min	up to 40
Speed, km/h	up to 79
Service ceiling, m	1200
Operating temperature range, °C	-10 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	10,000
Operating frequency range, MHz	915 / 5800
Manufacturer	Russia

FPV drone Katyusha

The FPV drone "Katyusha" is designed to destroy enemy manpower on the battlefield and in urban environments



Parameter name	Value
Maximum takeoff weight, kg	3.5
Payload mass, kg	up to 2.4
Range, km	up to 10
Flight time, min	10
Speed, km/h	up to 160
Practical ceiling, m	4000
Operating temperature range, °C	-20 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	868; 915 / 5850
Manufacturer	ANO Arkhangel Krasnodar

FPV drone "Product 1-390"

The FPV drone "Product 1-390" is designed for high-precision strikes against enemy armored vehicles and manpower with manual targeting via video signal.



Parameter name	Value
Maximum takeoff weight, kg	3.3
Payload weight, kg	up to 2
Range, km	up to 8
Flight time, min	up to 30
Speed, km/h	up to 120
Practical ceiling, m	20
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	N/A
Manufacturer	KBB-77 LLC Moscow

FPV drone "Slepen"

FPV drone "Slepen" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum take-off weight, kg	4.4
Payload, kg	up to 3.5
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 180
Service ceiling, m	800
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	868; 915 / 5850
Manufacturer	Russia

FPV drone "Ryabchik"

FPV drone Ryabchik is designed for strike and reconnaissance operations, escort assault groups



Parameter name	Value
Maximum take-off weight, kg	2
Payload, kg	up to 0.8
Range, km	up to 2
Flight time, min	up to 15
Speed, km/h	up to 70
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	868; 915 / 2400; 5800
Manufacturer	Russia

FPV drone "Courier"

The FRV-drone "Courier" is designed to engage enemy personnel , shelters, and lightly armored vehicles



Parameter name	Value
Maximum takeoff weight, kg	4.8
Payload, kg	up to 3.5
Range, km	up to 10
Flight time, min	up to 13
Speed, km/h	up to 60
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868; 915 / 2400; 5800
Manufacturer	Tula

FPV drone A8 3300 "Alesha"

The A8 3300 "Alesha" FPV drone is designed to destroy enemy manpower, armored vehicles, land and water transport, as well as to expose firing points and destroy artillery guns



Parameter name	Value
Maximum takeoff weight, kg	4.2
Payload, kg	up to 3
Range, km	up to 7
Flight time, min	up to 22
Speed, km/h	up to 120
Practical ceiling, m	3000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	520 / 5600
Manufacturer	IONOS LLC Rostov-on-Don

FPV drone "Boomerang-ERA"

The FPV drone "Boomerang-ERA" is designed to destroy enemy manpower and military equipment.



Parameter name	Value
Maximum take-off weight, kg	3.1
Payload, kg	up to 2
Range, km	up to 5
Flight time, min	up to 15
Speed, km/h	up to 200
Practical ceiling, m	1000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	868 / 5800
Manufacturer	VIT ERA and OOO OKB "Hover"

FPV drone "Granat-3"

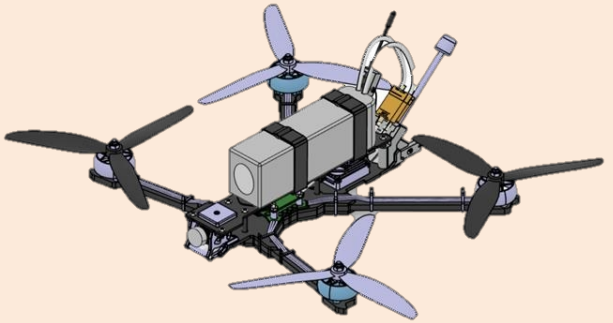
The Granat-3 FPV drone is designed to search for, detect, and recognize targets, guide weapons, and engage enemies on the battlefield, even in conditions of interference and lack of direct visibility.



Parameter name	Value
Maximum takeoff weight, kg	6
Payload, kg	up to 3
Range, km	up to 7
Flight time, min	up to 15
Speed, km/h	up to 35
Practical ceiling, m	3000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868; 915 / 2400; 5800
Manufacturer	JSC "OKB "Astron" Lytkarino

FPV drone "Monolith-FPV-7"

FPV drone "Monolith-FPV-7" is designed for striking enemy manpower and military equipment



Parameter name	Value
Maximum take-off weight, kg	0.654
Payload, kg	up to 3
Range, km	up to 10.9
Flight time, min	up to 15
Speed, km/h	up to 130
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4500 / 6000
Operating frequency range, MHz	868 – 915 / 5362 – 5945
Manufacturer	Monolith LLC Moscow

FPV drone "Osa"

FPV drone "Osa" is designed for strike and reconnaissance operations, accompanying assault groups



Parameter name	Value
Maximum takeoff weight, kg	6
Payload mass, kg	up to 2
Range, km	up to 4
Flight time, min	up to 25
Speed, km/h	up to 150
Service ceiling, m	400
Operating temperature range, °C	from -20 to +40 °C
Start/landing method	vertical
Battery capacity, mAh	4000
Operating frequency range, MHz	868–915 / 5800
Manufacturer	TSUGAM LLC

FPV drone "Loon-7 (10)"

The Loon-7 (10) FPV drone is designed to engage enemy personnel enemy forces and small shelters



Parameter name	Value
Maximum takeoff weight, kg	3.2
Payload weight, kg	up to 2.5
Range, km	up to 8
Flight time, min	up to 40
Speed, km/h	up to 140
Practical ceiling, m	4000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	7000
Operating frequency range, MHz	1080–1320
Manufacturer	NPO AVIABAS

FPV drone "Veter"

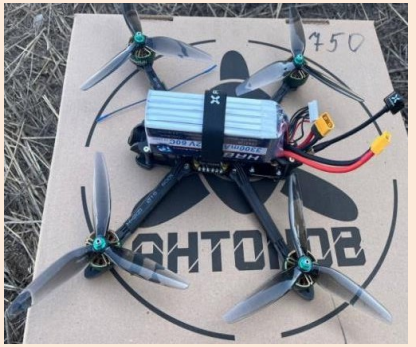
FPV drone "Veter" is designed for conducting aerial reconnaissance and striking enemy forces



Parameter name	Value
Maximum takeoff weight, kg	1.5
Payload mass, kg	up to 0.31
Range, km	up to 15
Flight time, min	up to 30
Speed, km/h	90
Service ceiling, m	3000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	1080–1320
Manufacturer	Kotlin Novator

FPV drone Antonov

FRV drone "Antonov" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum take-off weight, kg	5.2
Payload, kg	up to 4
Range, km	up to 7.5
Flight time, min	up to 15
Speed, km/h	up to 70
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	865; 915 / 5800
Manufacturer	Russia

9-inch FPV drone

FPV drone 9 inches is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum takeoff weight, kg	5.8
Payload, kg	up to 3
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 150
Practical ceiling, m	1600
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	7500
Operating frequency range, MHz	1080–1320
Manufacturer	Russia

10-inch FPV drone

FPV drone 10 inches is designed for strike and reconnaissance operations, escort actions assault teams



Parameter name	Value
Maximum takeoff weight, kg	5.8
Payload, kg	up to 3.5
Range, km	up to 10
Flight time, min	up to 15
Speed, km/h	up to 150
Practical ceiling height, m	1600
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	7500
Operating frequency range, MHz	1080–1320
Manufacturer	Russia

FPV drone "Komar"

The Komar FPV drone is designed for reconnaissance and destruction of enemy , and adjusting artillery fire



Parameter name	Value
Maximum takeoff weight, kg	1.8
Payload mass, kg	up to 1.5
Range, km	up to 10
Flight time, min	up to 30
Speed, km/h	up to 85
Practical ceiling, m	140
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5800
Operating frequency range, MHz	1080 – 1320
Manufacturer	Russia

FPV drone "Ultimatum"

The Ultimatum FPV drone is designed for reconnaissance, destruction enemy equipment , and adjusting artillery fire



Parameter name	Value
Maximum takeoff weight, kg	4.8
Payload, kg	up to 3
Range, km	up to 15
Flight time, min	up to 30
Speed, km/h	up to 200
Practical ceiling, m	180
Operating temperature range, °C	from -30 to +40
Start/landing method	vertical
Battery capacity, mAh	5800
Operating frequency range, MHz	1080–1320
Manufacturer	Russia

FPV drone "Piranha-7"

The Piranha FPV drone is designed for reconnaissance and destruction of enemy , and adjusting artillery fire



Parameter name	Value
Maximum takeoff weight, kg	3.8
Payload, kg	up to 2.5
Range, km	up to 7
Flight time, min	up to 13
Speed, km/h	up to 125
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	300; 735; 868; 915 / 1200; 4900; 5800
Manufacturer	SKB Piranha Yekaterinburg

FPV drone "Piranha-10"

FPV drone Piranha-10 is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum take-off weight, kg	6
Payload, kg	up to 4.5
Range, km	up to 11
Flight time, min	up to 13.3
Speed, km/h	up to 140
Service ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	300; 735; 868; 915 / 1200; 4900; 5800
Manufacturer	SKB Piranha Yekaterinburg

FPV drone "Lutik-Kamikaze"

The FPV drone "Lutik-Kamikaze" is designed for reconnaissance and delivery of ammunition to destroy manpower in dugouts and special shelters, as well as enemy armored vehicles.



Parameter name	Value
Maximum takeoff weight, kg	5.1
Payload, kg	up to 3.6
Range, km	up to 23
Flight time, min	up to 10
Speed, km/h	up to 130
Practical ceiling, m	1500
Operating temperature range, °C	-15 to +50
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	433; 868 / 1240; 5800
Manufacturer	Sakhalin Center Unmanned Technologies

FPV drone "Boomerang-10"

FPV drone "Boomerang-10" is designed for strike and reconnaissance operations, escort actions assault groups



Parameter name	Value
Maximum take-off weight, kg	6.5
Payload, kg	up to 3
Range, km	up to 6
Flight time, min	up to 20
Speed, km/h	up to 150
Practical ceiling, m	600
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	720-1020 / 4900-5900
Manufacturer	TSUGAM LLC

FPV drone "Type-450"

The Type-450 FPV drone is designed for reconnaissance, striking enemy manpower and military equipment



Parameter name	Value
Maximum take-off weight, kg	1.6
Payload weight, kg	up to 0.6
Range, km	up to 7
Flight time, min	up to 15
Speed, km/h	up to 130
Practical ceiling, m	100
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868–950 / 5333–5945
Manufacturer	RDR Group LLC Moscow

FPV drone "Microbe"

The Microbe FPV drone is designed for reconnaissance, destroying enemy equipment, and correcting artillery fire



Parameter name	Value
Maximum take-off weight, kg	4
Payload, kg	up to 2.5
Range, km	up to 15
Flight time, min	up to 16
Speed, km/h	up to 170
Service ceiling, m	300
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	4000
Operating frequency range, MHz	503; 750; 866 / 1100 – 1300; 4900
Manufacturer	Ploshchad

FPV drone "Dozor"

FPV drone "Dozor" is designed for conducting reconnaissance, and to destroy enemy manpower and military equipment.



Parameter name	Value
Maximum takeoff weight, kg	2.7
Payload, kg	up to 1.2
Range, km	up to 10
Flight time, min	up to 20
Speed, km/h	up to 130
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	868–915 / 5800
Manufacturer	Prom Composite LLC Moscow

FPV drone "Upyry"

FPV drone "Upyry" is designed for destroying enemy personnel enemy forces and equipment



Parameter name	Value
Maximum take-off weight, kg	3.5
Payload, kg	up to 1.5
Range, km	up to 7
Flight time, min	up to 30
Speed, km/h	up to 120
Practical ceiling, m	1000
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	860–1020/1200; 5800
Manufacturer	UralDronZavod LLC Yekaterinburg

Section 2

FPV DRONES OF THE AIRCRAFT AND HELICOPTER TYPE

- FPV drone "Chernika-1"
- FPV drone "Chernika-2"
- FPV drone "Priviet-82"
- FPV drone "Komar"
- FPV drone "Molniya-1"
- FPV drone "Molniya-1 AI"

FPV drone "Chernika-1"

The Chernika-1 aircraft-type FPV drone is designed for high-precision strikes against enemy armored vehicles and personnel, with manual targeting via video signal.



FPV drone "Chernika-2"

The Chernika-2 aircraft-type FPV drone is designed for high-precision strikes against enemy armored vehicles and manpower, with manual targeting via video signal.



Parameter name	Value
Maximum take-off weight, kg	5.8
Payload, kg	up to 2.2
Range, km	up to 25
Flight time, min	up to 80
Speed, km/h	up to 120
Practical ceiling, m	1000
Operating temperature range, °C	from -10 to +40
Launch/landing method	pneumatic catapult
Battery capacity, mAh	5000
Operating frequency range, MHz	740; 868; 915 / 1200; 2400; 5800
Manufacturer	Russia

Parameter name	Value
Maximum takeoff weight, kg	5.8
Payload, kg	up to 2.2
Range, km	up to 25
Flight time, min	up to 80
Speed, km/h	up to 120
Practical ceiling, m	1000
Operating temperature range, °C	from -10 to +40
Launch/landing method	pneumatic catapult
Battery capacity, mAh	5000
Operating frequency range, MHz	740; 868; 915 / 1200; 2400; 5800
Manufacturer	Russia

FPV drone "Privet-82"

The FPV drone "Privet-82" is designed for high-precision strikes against enemy air defense systems and manpower with manual targeting via video signal.



Parameter name	Value
Maximum take-off weight, kg	8.5
Payload weight, kg	up to 4.5
Range, km	up to 30
Flight time, min	up to 20
Speed, km/h	up to 170
Service ceiling, m	2000
Operating temperature range, °C	from -10 to +40
Start/landing method	vertical
Battery capacity, mAh	8000
Operating frequency range, MHz	930 / 2400; 5800
Manufacturer	SPDBC LLC

FPV drone "Komar"

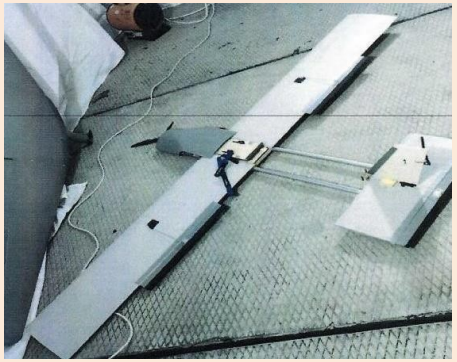
The FRV drone of the helicopter type "Komar" is designed for high-precision strikes against enemy armored vehicles and manpower, with manual targeting via video signal.



Parameter name	Value
Maximum take-off weight, kg	1.8
Payload weight, kg	up to 0.5
Range, km	up to 10
Flight time, min	up to 30
Speed, km/h	up to 85
Practical ceiling, m	up to 140
Operating temperature range, °C	from -20 to +40
Start/landing method	vertical
Battery capacity, mAh	5800
Operating frequency range, MHz	1080–1320
Manufacturer	Russia

FPV drone "Molniya-1"

The Molniya-1 FPV drone is designed for high-precision strikes against enemy air defense systems and manpower, with manual targeting via video signal.



Parameter name	Value
Maximum take-off weight, kg	5
Payload weight, kg	up to 3.5
Range, km	up to 40
Flight time, min	up to 40
Speed, km/h	up to 120
Practical ceiling, m	1000
Operating temperature range, °C	from -10 to +40
Launch/landing method	pneumatic catapult
Battery capacity, mAh	4000
Operating frequency range, MHz	915 / 5800
Manufacturer	AtlantAero LLC Taganrog

FPV drone "Molniya-1 AI"

The FPV drone "Molniya-1 AI" is designed for high-precision strikes against enemy airborne combat vehicles and manpower, with manual targeting via video signal.



Parameter name	Value
Maximum takeoff weight, kg	5
Payload, kg	up to 3.5
Range, km	up to 40
Flight time, min	up to 40
Speed, km/h	up to 120
Practical ceiling, m	1500
Operating temperature range, °C	from -10 to +40
Launch/landing method	pneumatic catapult
Battery capacity, mAh	5000
Operating frequency range, MHz	915 / 5800
Manufacturer	AtlantAero LLC Taganrog

Section 3

COPTER-TYPE UAV WITH FPV FUNCTION

- UAV-KT "DJI Mavic 3 Classic"
- UAV-KT "DJI Mavic 3T"
- DJI Mavic 2 Enterprise Advanced UAV-KT
- AUTEL EVO UAV-KT
- DJI Matrice 300 RTK UAV-KT
- UAV-KT "DJI Air 2S"
- UAV-KT "DJI Mini 2"
- UAV-KT DJI PHANTOM 3
- DJI 6S UAV-KT with Liana-3 ground power station
- Sturm software and hardware complex
- UAV "Dji Mavic Pro Platinum"
- ZALA 421-22 UAV-KT
- Sych UAV-KT
- Grifon-41 UAV
- UAV-KT "Liza"
- Volk UAV-KT
- Sibir-1 complex
- Kvazimachta complex

DJI Mavic 3 Classic UAV-KT

The DJI Mavic 3 Classic UAV-KT is designed for aerial reconnaissance with information transmission via Wi-Fi radio data transmission channel, as well as for striking enemy manpower and military equipment using F-1, RKG-3, RGD/RGN, 0.5-OFS, and VOG-17 drop devices.



Parameter name	Value
Maximum take-off weight, kg	1.6
Payload weight, kg	up to 0.6
Range, km	up to 15
Flight time, min	up to 46
Speed, km/h	up to 68
Service ceiling, m	6000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT "DJI Mavic 3T"

The DJI Mavic 3T UAV-KT is designed for aerial reconnaissance with information transmission via Wi-Fi radio data transmission channel, as well as for striking enemy manpower and military equipment using F-1, RKG-3, RGD/RGN, 0.5-OFS, and VOG-17 drop devices.



Parameter name	Value
Maximum takeoff weight, kg	1.6
Payload mass, kg	up to 0.6
Range, km	up to 15
Flight time, min	up to 45
Speed, km/h	up to 75.6
Practical ceiling height, m	5000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-CT "DJI Mavic 2 Enterprise Advanced

The DJI Mavic 2 Enterprise Advanced UAV-KT is designed to conduct aerial reconnaissance via Wi-Fi data transmission, as well as to destroy enemy manpower and military equipment using F-1, RKG-3, RGD/RGN, 0.5-OFSPA, VOG-17



Parameter name	Value
Maximum take-off weight, kg	1.4
Payload weight, kg	up to 0.5
Range, km	up to 8
Flight time, min	up to 31
Speed, km/h	up to 72
Service ceiling, m	6000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	3950
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT "AUTEL EVO"

The AUTEL EVO UAV-KT is designed for aerial reconnaissance with information transmission via Wi-Fi.



Parameter name	Value
Maximum takeoff weight, kg	1.6
Payload weight, kg	up to 0.5
Range, km	up to 8
Flight time, min	up to 31
Speed, km/h	up to 72
Service ceiling, m	6000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	3500
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT "DJI Matrice 300 RTK"

The DJI Matrice 300 RTK UAV-KT is designed for aerial reconnaissance with information transmission via radio channels data Wi-Fi, and as well as strikes live enemy forces and military equipment using F-1, RKG-3, RGD/RGN, 0.5-OFS, VOG-17



Parameter name	Value
Maximum takeoff weight, kg	9
Payload weight, kg	up to 3
Range, km	up to 8
Flight time, min	up to 55
Speed, km/h	up to 83
Service ceiling, m	7000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	5935
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT DJI Air 2S

The DJI Air 2S UAV is designed for aerial reconnaissance reconnaissance with information transmission via Wi-Fi.



Parameter name	Value
Maximum takeoff weight, kg	0.6
Payload mass, kg	up to 0.15
Range, km	up to 12
Flight time, min	up to 31
Speed, km/h	up to 68
Service ceiling, m	5000
Operating temperature range, °C	-10 to +40
Takeoff/landing method	Vertical
Battery capacity, mAh	3500
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT "DJI Mini 2"

The DJI Mini 2 UAV-KT is designed for aerial reconnaissance with information transmission via a Wi-Fi radio data channel.



Parameter name	Value
Maximum take-off weight, kg	0.25
Payload weight, kg	up to 0.15
Range, km	up to 10
Flight time, min	up to 30
Speed, km/h	up to 57
Practical ceiling, m	5000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	3950
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT DJI PHANTOM 3

The DJI PHANTOM 3 UAV-KT is designed for aerial reconnaissance with information transmission via a radio channel data Wi-Fi, and as well as striking live enemy forces and military equipment using F-1, RKG-3, RGD/RGN, 0.5-OFS, and VOG-17 release devices.



Parameter name	Value
Maximum takeoff weight, kg	3
Payload, kg	up to 1.2
Range, km	up to 8
Flight time, min	up to 25
Speed, km/h	up to 57
Service ceiling, m	5000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	3500
Operating frequency range, MHz	2400 / 5100
Manufacturer	China

UAV-KT "DJI 6S" with ground Liana-3 power supply station

The DJI 6S UAV-KT with the Liana-3 ground power station is designed for reconnaissance and surveillance, using a power cable and LAN to transmit video and control signals



Parameter name	Value
Maximum take-off weight, kg	20
Payload weight, kg	up to 15
Range, km	up to 20
Flight time, min	up to 72
Speed, km/h	up to 30
Service ceiling, m	100
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	-
Operating frequency range, MHz	n/a
Manufacturer	China

Software and hardware complex "Sturm"

The Sturm software and hardware complex is designed for operating an FPV drone during combat operations without FPV control skills.



Parameter name	Value
Maximum takeoff weight, kg	3.0
Payload, kg	up to 2.3
Range, km	up to 7
Flight time, min	up to 15
Speed, km/h	up to 80
Practical ceiling, m	5000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	900 / 5200; 5800
Manufacturer	China

UAV "Dji Mavic Pro Platinum"

The Dji Mavic Pro Platinum UAV is designed for aerial reconnaissance with information transmission via a Wi-Fi data channel.

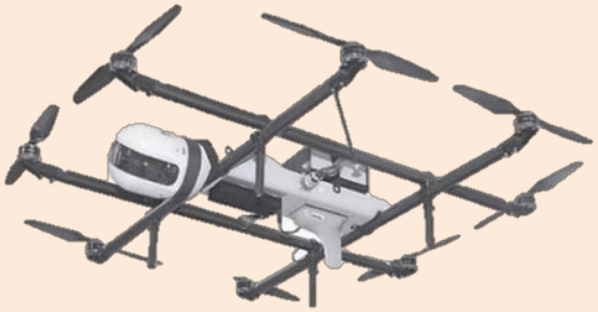
The UAV flies in semi-automatic or automatic modes, including tracking selected moving objects without operator intervention.



Parameter name	Value
Maximum take-off weight, kg	2.2
Payload weight, kg	up to 0.3
Range, km	up to 7
Flight time, min	up to 30
Speed, km/h	up to 65
Practical ceiling, m	5000
Operating temperature range, °C	0 to +40
Start/landing method	vertical
Battery capacity, mAh	3850
Operating frequency range, MHz	900 / 5200; 5800
Manufacturer	China

ZALA 421-22 UAV-KT

The ZALA 421-22 UAV-KT is designed for reconnaissance and surveillance, video signal transmission, and control signals.



Parameter name	Value
Maximum takeoff weight, kg	8
Payload, kg	up to 5
Range, km	up to 5
Flight time, min	up to 35
Speed, km/h	up to 30
Practical ceiling, m	1000
Operating temperature range, °C	-10 to +40
Start/landing method	vertical
Battery capacity, mAh	5000
Operating frequency range, MHz	N/A
Manufacturer	Russia

UAV-KT "Sych"

UAV-KT "Sych" is designed for counterbattery counter-battery tasks



Parameter name	Value
Maximum take-off weight, kg	8
Payload weight, kg	up to 5
Range, km	up to 35
Flight time, min	up to 30
Speed, km/h	up to 70
Practical ceiling, m	180
Operating temperature range, °C	from -30 to +50
Start/landing method	vertical
Battery capacity, mAh	4500
Operating frequency range, MHz	N/A
Manufacturer	STC LLC St. Petersburg

UAV "Griffon-41"

The Gryphon-41 UAV is designed to increase radio communication range (up to 25 km) in tactical command and control.



Parameter name	Value
Maximum take-off weight, kg	6
Payload, kg	up to 4
Range, km	up to 5
Flight time, min	up to 540
Speed, km/h	up to 70
Service ceiling, m	500
Operating temperature range, °C	from -30 to +50
Start/landing method	vertical
Battery capacity, mAh	220 V / 100000 W 220 V / 10,000
Operating frequency range, MHz	N/A
Manufacturer	Plaz LLC, St. Petersburg

UAV-KT Lisa

The UAV-KT "Lisa" is designed for visual reconnaissance, detection and recognition of targets, and destruction of targets at speeds of up to 60 km/h as its main mode of operation.

Means of destruction RGD-5, RG-42, F-1, RGN/RGO, RKG-3 -

ammunition supply 8 pcs.



Parameter name	Value
Maximum takeoff weight, kg	8
Payload mass, kg	up to 3
Range, km	up to 8
Flight time, min	up to 25
Speed, km/h	up to 60
Practical ceiling, m	4000
Operating temperature range, °C	-20 to +40
Start/landing method	vertical
Battery capacity, mAh	6000
Operating frequency range, MHz	N/A
Manufacturer	PromComposite LLC Moscow

UAV-KT "Wolf"

The BpLA-KT "Wolf" is designed to combat enemy UAVs using a Kevlar net



Parameter name	Value
Maximum take-off weight, kg	6.5
Payload, kg	up to 1.6
Range, km	up to 4
Flight time, min	up to 28
Speed, km/h	up to 80
Practical ceiling, m	1000
Operating temperature range, °C	-20 to +40
Start/landing method	vertical
Battery capacity, mAh	5800
Operating frequency range, MHz	N/A
Manufacturer	PromKompozit LLC Moscow

Sibir-1 Complex

The Sibir-1 complex is designed to increase the range of radio communication networks deployed in tactical command and control.





Parameter name	Value
Maximum take-off weight, kg	8.5
Payload mass, kg	up to 5
Range, km	up to 2
Flight time, min	up to 40
Speed, km/h	up to 36
Practical ceiling height, m	500
Operating temperature range, °C	from -30 to +50
Start/landing method	vertical
Battery capacity, mAh	5800
Operating frequency range, MHz	N/A
Manufacturer	Gaskar Group LLC Moscow

Quasi-mast complex

The Quasi-Mast complex is designed to increase the range of radio communications (up to 15 km) in the tactical command link





Parameter name	Value
Maximum take-off weight, kg	6.7
Payload, kg	up to 3
Range, km	up to 3
Flight time, min	up to 40
Speed, km/h	up to 27
Service ceiling, m	100
Operating temperature range, °C	from -30 to +50
Start/landing method	vertical
Battery capacity, mAh	6200
Operating frequency range, MHz	N/A
Manufacturer	YUVS Avia LLC Moscow

Section 4

IMPACT LOAD OF FPV DRONES

- 0.5 kg high-explosive fragmentation warhead
- 0.8 kg high-explosive fragmentation warhead
- 1.7 kg high-explosive fragmentation warhead
- 2.5 kg high-explosive fragmentation warhead
- 1.4 kg fragmentation-cluster warhead
- Warhead: cumulative-incendiary, caliber 0.5 kg
- 3 kg high-explosive fragmentation warhead
- 1.4 kg cumulative-fragmentation-incendiary warhead
- 0.5 kg cumulative incendiary warhead

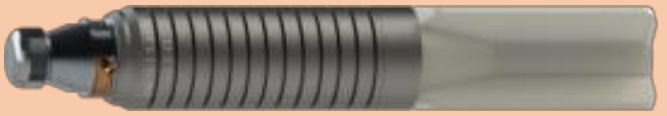
- PG-7V warhead
- KZ-6 cumulative charge
- TBG-7V warhead
- F-1 grenade
- RGN grenade
- RGO grenade
- 82 mm mine
- Kaplya weapon
- Shayba weapon

0.5 kg high-explosive fragmentation warhead



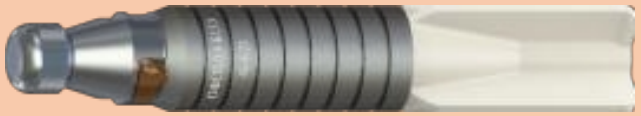
Parameter name	Value
Designation	OFSP-0.5-B133
Type of placement	Dumpable
Type of action	Fragmentation-high explosive
Weight, kg	0.5
Dimensions, mm	Ø48 x 188
Fuse type	B-133
Status	Serial production

Warhead High-explosive fragmentation warhead, 1.7 kg



Parameter name	Value
Designation	OFSP-1.7-A133 / OFSP-1.7-B133
Type of placement	Dumpable
Type of action	Fragmentation-high explosive
Weight, kg	1.7
Dimensions, mm	Ø55 x 352
Fuse type	A-133 / B-133
Status	Serial production

Warhead 0.8 kg high-explosive fragmentation warhead



Parameter name	Value
Designation	OFSP-0.8-B133 / OFSP-0.8-A133
Type of placement	Dischargeable
Type of action	Fragmentation-high explosive
Weight, kg	0.8
Dimensions, mm	Ø48 x 231
Fuse type	B-133 / A-133
Status	Serial production

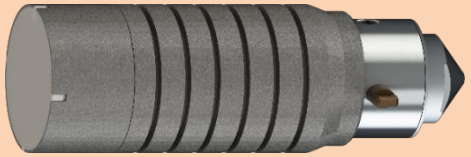
Warhead 2.5 kg high-explosive fragmentation warhead



Parameter name	Value
Designation	OFSP-2.5-A133 / OFSP-2.5-B133
Type of placement	Dumpable
Type of action	Fragmentation-high explosive
Weight, kg	2.5
Dimensions, mm	Ø60 x 399
Fuse type	A-133 / B-133
Status	Serial production

Weapons for FPV drones

1.4 kg fragmentation warhead



Parameter name	Value
Designation	OPBCH-1.4-A795
Type of placement	Insert
Type of action	Fragmentation-cluster
Weight, kg	1.4
Dimensions, mm	Ø60 x 147
Fuse type	A-795
Status	development, testing

Warhead High-explosive fragmentation warhead, 3 kg caliber



Parameter name	Value
Designation	OFPBCH-3Sh-A795
Type of accommodation	Deposit
Type of action	Fragmentation-high-explosive-cluster
Weight, kg	3.0
Dimensions, mm	Ø58 x 267
Fuse type	A-795
Status	development, testing

Warhead 0.5 kg cumulative incendiary



Parameter name	Value
Designation	KZSP-0.5-A688
Type of placement	Droppable
Type of action	Cumulative-incendiary
Weight, kg	0.5
Dimensions, mm	Ø48 x 214
Fuse type	A-688
Status	Serial production

Warhead 1.4 kg cumulative-fragmentation-incendiary



Parameter name	Value
Designation	KOZBC-1.4-A795
Type of placement	Insert
Type of action	Cumulative-fragmentation-incendiary
Weight, kg	1.4
Dimensions, mm	Ø70 x 160
Fuse type	A-795
Status	development, testing

Weapons for FPV drones

Warhead: cumulative incendiary, caliber 0.5 kg



Parameter name	Value
Designation	OPBCH-1.4-A795
Type of placement	Insert
Type of action	fragmentation-bundle
Weight, kg	1.4
Dimensions, mm	Ø60 x 147
Fuse type	A-795
Status	development, testing

Cumulative charge KZ-6



Parameter name	Value
Designation	KZ-6
Type of placement	Insert
Type of action	Cumulative
Weight, kg	3
Dimensions, mm	Ø112×300
Fuse type	Instantaneous electric detonator (EDP-R)
Status	Serial production

Warhead PG-7V



Parameter name	Value
Designation	PG-7V
Type of placement	Insert
Type of action	Cumulative
Weight, kg	2.2
Dimensions, mm	Ø85×270
Fuse type	piezoelectric VP-7
Status	Serial production

Warhead TBG-7V



Parameter name	Value
Designation	TBG-7V
Type of placement	Insert
Type of action	thermobaric
Weight, kg	4.5
Dimensions, mm	Ø105×310
Fuse type	piezoelectric VP-7
Status	Serial production

Weapons for FPV drones

F-1 grenade



Parameter name	Value
Designation	
Type of placement	Insert
Action type	Fragmentation
Weight, kg	0.6
Dimensions, mm	Ø60
Fuse type	UDZ
Status	Serial production

RGO grenade



Parameter name	Value
Designation	(7G22)
Type of placement	Insert
Type of action	Fragmentation
Weight, kg	0.52
Dimensions, mm	Ø60
Fuse type	UDZ
Status	Serial production

RGN grenade



Parameter name	Value
Designation	(7G21)
Type of placement	Insert
Type of action	Fragmentation
Weight, kg	0.29
Dimensions, mm	Ø60
Fuse type	UDZ
Status	Serial production

82 mm mine



Parameter name	Value
Designation	3VO18 shot with 3O-12 mine
Type of placement	Insertable
Type of action	Fragmentation
Weight, kg	3.1
Dimensions, mm	Ø82×310
Fuse type	M-5 (M-6)
Status	Serial production

Weapons for FPV drones

Weapon "Kaplya"

The Kaplya weapon is designed to destroy lightly armored vehicles when used with FPV UAVs.



Parameter name	Value
Manufacturer	RFNC VNIIEF Moscow
Type of placement	Suspended
Armor penetration, mm	40
Weight, kg	2.3
Warhead caliber, mm	140
Warhead type	Impact core
Status	development, testing

Weapon "Shayba"

The Shayba weapon is designed to destroy enemy personnel when used with FPV UAVs.



Parameter name	Value
Manufacturer	RFNC VNIIEF Moscow
Type of accommodation	Suspended
Effective range, m	40
Weight, kg	1.9
Warhead caliber, mm	96
Warhead type	High-explosive fragmentation
Status	development, testing

Section 5

SUPPRESSION MEANS OF FPV DRONES

- Pishchal-PRO portable complex
- Garpun-2M2 portable complex
- Stillet portable complex
- Block portable complex
- Phantom portable complex
- Prascha-01 portable UAV countermeasure system
- Argument portable complex
- Patrol wearable complex
- LP-801 portable complex
- Arkan-2 UAV jammer
- SPD-2 portable complex
- Zhezl-2 UAV jammer
- SHUM-22TK control channel suppression system
- Taran-PRO portable complex
- Strizh-3 UAV detection and protection system
- Argus-Antifuria mobile electronic warfare station
- Strizh portable system
- PARS-PR Plombir
- Patrol mobile complex
- PARS-F Filya
- PARS-FB (Filia onboard)
- SERP-VS6 UAV jammer
- Sfera-10 portable complex
- Pereod UAV jammer
- UAV jammer "Grozha" "Bronya"
- Volnorez UAV control channel jammer
- Kapyushon K8 UAV jammer
- FPV drone jamming device LGSh-689

Means of countering UAVs

Pishchal-PRO portable system



The Pishchal-PRO portable complex is designed to disrupt UAV flight by suppressing communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	433; 915; 1500; 2400; 5800
Autonomy, min	up to 60
Range, m	up to 2000
Built-in battery capacity, Ah	10
Dimensions, cm	200 x 240 x 930
Weight, kg	3.2

Stillet portable complex



The Stiletto portable complex is designed to disrupt the flight by suppressing communication, control, and navigation channels

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	390–490; 850–1400; 2200–2400
Autonomy, min	up to 40
Range, m	up to 1000
Built-in battery capacity, Ah	-
Dimensions, cm	-
Weight, kg	9

Harpun-2M2 portable system



The Harpoon-2M2 portable complex is designed to disrupt the flight mission of UAVs by suppressing communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	433; 868; 1575 / 1602; 2400; 5800
Autonomy, min	up to 60
Range, m	-
Built-in battery capacity, Ah	-
Dimensions, cm	124 x 31.5 x 19
Weight, kg	9.5

Block portable system

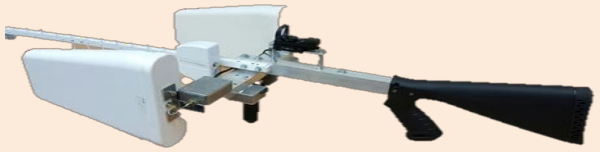


The Block wearable complex is designed to disrupt UAV flight missions by suppressing communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	433; 915; 1500; 2400; 5800
Autonomy, min	up to 60
Range, m	from 1.5 to 2.3
Built-in battery capacity, Ah	10
Dimensions, cm	101 x 30 x 14
Weight, kg	4.5

Means of countering UAVs

Phantom portable complex



The Phantom portable complex is designed to suppress channels UAV communications, control, and navigation

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Frequencies suppressed, MHz	868–930; 1570–1607; 2400
Autonomy, min	up to 60
Range, m	up to 300
Built-in battery capacity, Ah	-
Dimensions, cm	-
Weight, kg	8

Argument portable system

The Argument wearable system is designed to suppress UAV communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Frequencies suppressed, MHz	430–440; 860–930; 1560–1650; 2400 – 2500; 5180 – 5350
Autonomy, min	up to 60
Range, m	up to 2100
Built-in battery capacity, Ah	-
Dimensions, cm	-
Weight, kg	6

Prascha-01 portable anti-UAV system



The Pracha-01 system is designed to suppress UAV communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	400–5800
Autonomy, min	60
Range, m	up to 2500
Suppression sector, degrees	20
Dimensions, cm	260 x 900 x 140
Weight, kg	7

Patrol portable complex



The Patrol portable system is designed to suppress UAV communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	2400; 5800
Autonomy, min	up to 240
Range, m	up to 500
Suppression sector, degrees	16
Dimensions, cm	260 x 900 x 140
Weight, kg	5.5

Means of countering UAVs

LPD-801 portable system



The LPD-801 portable complex is designed to suppress the communication, control, and navigation channels of UAVs.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Suppressed communication frequencies, GHz	1.1–1.8, 2.4, 5.8
Autonomy, min	30
Range, m	1000
Built-in battery capacity, Ah	5000
Weight, kg	6

SPD-2 portable system



The SPD-2 portable complex is designed to suppress the communication, control, and navigation channels of UAVs.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Suppressed communication frequencies, GHz	0.430-0.440; 0.860-0.930; 1.560-1650; 2.400-2.500; 5.180-5.350
Autonomy, min	300
Range, m	up to 1000
Weight, kg	6

Arkan-2 UAV jammer



The Arkan-2 UAV jammer is designed to suppress UAV communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Suppressed communication frequencies, GHz	0.430-0.440; 0.860-0.930; 1.560-1650; 2.400-2.500; 5.180-5.350
Autonomy, min	30
Range, m	up to 1
Weight, kg	-

Zhezl-2 UAV jammer



The Zhezl-2 UAV jammer is designed to suppress UAV communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Suppressed communication frequencies, MHz	1100-1500; 1600-5800
Autonomy, min	60
Range, m	up to 500
Suppression sector	-
Weight, kg	7

SHUM-22TK control and navigation channel suppression system



The SHUM-22TK complex is designed to counteract the communication channels of navigation systems and the control channels of unmanned aerial vehicles.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Suppressed communication frequencies, MHz	400–470; 860–930 2400 – 2483.5; 5150 – 5945; 1176–1228; 1242–1279; 1575.32 – 1575.52; 1598 – 1606
Range, m	up to 1500
Dimensions, cm	62 x 55 x 33
Weight, kg	36

Strizh-3 UAV Detection and Protection System



The Strizh-3 UAV detection and protection system is designed to detect unmanned aerial vehicles and block their control and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Frequencies suppressed, MHz	433, 868, 900, 1200, 2400, 5200, 5800
Range, m	up to 2500
Built-in battery capacity, Ah	-
Dimensions, cm	35 x 35 x 170
Weight, kg	15

Taran-PRO portable complex



The Taran-PRO portable complex is designed to disrupt the flight mission of UAVs by suppressing communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	340–460; 824–980; 1200–1500; 2200–2458; 4000 – 5250
Autonomy, min	-
Range, m	up to 2600
Suppression sector, degrees	360
Weight, kg	-

Argus-Antifuria mobile electronic warfare station



The Argus-Antifuria mobile electronic warfare station is designed to prevent unmanned aerial vehicles from flying in the area of operation by generating high-power directional interference.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	400; 900; 1200; 1500; 2400; 5800
Autonomy, min	-
Range, m	2000
Suppression sector, degrees	360
Dimensions, cm	150 x 120 x 120
Weight, kg	18

Means of countering UAVs

Strizh portable complex



The Strizh portable complex is designed for automatic detection of control signals and jamming of detected signals or navigation signals.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS
Frequencies suppressed, MHz	433; 868; 900; 1575; 1600; 2400; 5800
Range, m	up to 1000
Suppression sector, hail	360
Dimensions, cm	35 x 35 x 170
Weight, kg	46

Mobile complex "Patrol"



The Patrol mobile complex is designed for continuous radio surveillance, detection of signals from UAVs, and generation of countermeasures in the protected area.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, BeiDou, Galileo (L1, E1, G1)
Frequencies suppressed, MHz	400–6000, Wi-Fi, LTE
Range, m	up to 25,000
Suppression sector, hail	360

PARS-PR "Plombir"



PARS-PR "Plombir" is designed for stationary protection of local facilities and vehicles from UAVs and FPV drones

Parameter name	Value
Range, m	up to 2000
Radiated power, W	up to 80
Suppressed frequencies, MHz	430–470; 840–910; 925–960; 1160–1280; 1570–1610; 2400–2500; 2500 – 2600; 5715 – 5870
Suppression sector, degrees	30 – 80
Weight, kg	20

PARS-F "Filia"



PARS-F "Filia" is designed to protect local facilities and vehicles from FPV drones

Parameter name	Value
Radiated power, W	up to 20
Suppressed frequencies, MHz	820–960; 1160–1280
Autonomy, min	up to 240
Range, m	up to 300
Suppression sector, degrees	360
Dimensions, cm	300 x 200 x 200
Weight, kg	5

Means of countering UAVs

PARS-FB ("Fila Bortovaya")



PARS-FB "Filia Bortovaya" is designed to protect vehicles from FPV drones

Parameter name	Value
Radiated power, W	up to 20
Suppressed frequencies, MHz	820–960; 1160–1280
Autonomy, min	up to 240
Range, m	up to 300
Suppression sector, degrees	360
Dimensions, cm	20 x 10 x 20
Weight, kg	1

Sfera-10 portable complex



The Sfera-10 portable complex is designed to disrupt the flight mission of UAVs by suppressing communication, control, and navigation channels.

Parameter name	Value
Suppressed navigation systems	GPS, GLONASS, Galileo, BeiDou
Frequencies suppressed, MHz	400 to 6000; Wi-Fi; LTE
Autonomy, min	-
Range, m	3000
Suppression sector, degrees	360
Dimensions, cm	101 (diameter), 20 (height)
Weight, kg	15

SERP-VS6 UAV jammer



SERP-VS6 is designed to suppress UAV control channels and disrupt the operation of onboard receivers of global navigation satellite systems at a distance of up to 5 km.

Parameter name	Value
Range, m	up to 5000
Radiated power, W	up to 80
Suppressed frequencies, MHz	430-470; 840-910; 925-960; 1160-1280; 1570-1610; 2400-2500; 2500-2600; 5715-5870
Suppression sector, degrees	360
Weight, kg	10

Pereod UAV jammer



The Pereod jammer is designed to combat unmanned aerial vehicles.

Parameter name	Value
Radiated power, W	up to 20
Suppressed frequencies, MHz	920; 1200; 1500; 2400; 5800
Autonomy, min	up to 40
Range, m	up to 300
Suppression sector, degrees	75
Dimensions, cm	n/a
Weight, kg	7

Means of countering UAVs

Groz-Bronya UAV jammer



The Groza Bronya UAV jammer is designed to be installed on vehicles and armored vehicles to protect against FPV drones in a 360-degree sector. It consists of eight jamming modules and a power supply unit.

Parameter name	Value
Radiated power, W	up to 20
Suppressed communication frequencies, MHz	400–500; 630–780; 770–900; 890–1020; 1150–1300; 2400–2500; 5100 – 5300; 5700 – 5900
Autonomy, min	-
Range, m	from 50 m
Suppression sector	360 / 90
Dimensions, cm	350 x 350 x 240
Weight, kg	16

Kapyushon K8 UAV jammer



UAV jammer " K8 Hood" is designed to protect against strike unmanned aerial vehicles, FPV drones, and drones with drop-offs

Parameter name	Value
Radiated power, W	up to 20
Suppressed communication frequencies, MHz	400–500; 630–780; 770–900; 890–1020; 1150–1300; 2400–2500; 5100 – 5300; 5700 – 5900
Autonomy, min	up to 240
Range, m	up to 500
Suppression sector, degrees	16
Dimensions, cm	n/a
Weight, kg	5.5

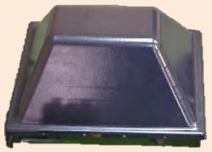
UAV control channel suppression device "Volnorez"



Device for suppression control channels control of FPV drones "Wavebreaker" is designed for protect weapons, military and special equipment from damage by FPV drones

Parameter name	Value
Radiated power, W	up to 20
Suppressed frequencies, MHz	863–886; 902–928
Autonomy, min	-
Range, m	up to 300
Suppression sector	Circular
Dimensions, cm	300 x 200
Weight, kg	8

FPV drone jamming device "LGSh-689"



Device for jamming FPV drone control channels
The LGSh-689 is designed to protect weapons, military and special equipment from being targeted by FPV drones.

Parameter name	Value
Radiated power, W	up to 20
Suppressed frequencies, MHz	863–886; 902–928
Autonomy, min	-
Range, m	up to 300
Suppression sector	Circular
Dimensions, cm	300 x 300 x 200
Weight, kg	10

A

AUTEL EVO 38

D

DJI 6S 41

DJI Air 2S 39

DJI Matrice 300 RTK 39

DJI Mavic 2 Enterprise Advanced 38

DJI Mavic 3 Classic 37

DJI Mavic 3T 37

Dji Mavic Pro Platinum 42

DJI Mini 2 40

DJI PHANTOM 3 40

F

5-inch FPV drone 7

7-inch FPV drone 7

9-inch FPV drone 26

10-inch FPV drone 26

X

XL-10 11

Z

ZALA 421-22 42

A

Alyosha 22

Antonov 25

Argument 54

Argus-Antiphury 56

Arcanum-2 55

ASK X-2 13

B

Block 53

0.5 kg cumulative incendiary warhead 48

0.5 kg cumulative incendiary warhead 49

0.5 kg high-explosive fragmentation warhead 47

0.8 kg high-explosive fragmentation warhead 47

1.4 kg cumulative-fragmentation-incendiary warhead 48

1.4 kg fragmentation-cluster warhead 48

1.7 kg high-explosive fragmentation warhead 47

High-explosive fragmentation warhead, 2.5 kg caliber 47

3 kg high-explosive fragmentation warhead 48

PG-7V warhead 49

TBG-7V warhead 49

Brus-1 9

Boomerang 8

Boomerang-10 29

Boomerang-E 9

Boomerang-ERA 22

B

Veles 14

Vespa 16

Wind 25

Wolf 44

Breakwater 59

G

Harpun-2M2 53

Hortensia-7 10

Hydrangea-10 10

Pomegranate-3 23

RGN grenade 50

RGO grenade 50

F-1 grenade 50

Grozha 59

Griffon-41 43

D

D-11-1 15

Darwin X9 8

Dozor 31

Z

Staff-2 55

Zhmil-17 16

Z

KZ-6 charge..... 49

I

Product 1-390..... 20

K

Drop... 51
… … … …

U

Ultimatum..... 27

Ghoul..... 31

F

Phantom..... 54

Filia..... 57

Side fillet..... 58

X

Chimera..... 12

Predator-2PG..... 19

Ч

Blueberry-1..... 33

Chernika-2..... 33

Sh

Washer..... 51

Bumblebee..... 14

Storm..... 41

SHUM-22TK..... 56

This reference guide has been developed based on the final reports of military units on the use of FPV drones and countermeasures against them in special military operations conducted by the Armed Forces of the Russian Federation.

Please send any comments or suggestions regarding this reference book to the department Combat Experience Summary Department of the Joint Group of Forces.

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